

Vidhi Jain

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Education

Ph.D. in Robotics from Carnegie Mellon University (Robotics Institute).

Robotics Institute, Advisor: [Yonatan Bisk](#)

Masters of Science in Robotics, Carnegie Mellon University.

Robotics Institute, GPA 4.0 Advisor: [Katia Sycara](#)

Bachelor of Engineering (Honors) in Computer Science, BITS Pilani.

CGPA 9.06/10.00, Distinction, Advisor (off-campus thesis): [Aaron Courville](#)

Experience

- Jul'25– **OpenAI, San Francisco, CA, USA.**
- Mar'25 Researcher in AI transformation team with [Boris Power](#)
- Drove the capability improvement for frontier model for specification to code generation tasks, utilizing reasoning models and verifiable rewards.
 - Developed synthetic data pipelines to convert code into domain specific language for better verification and preventing reward hacking behaviors in RL post-training.
- Jun'23– **Google DeepMind, Mountain View, CA, USA.**
- Feb'24 Student Researcher with [Debidatta Dwibedi](#)
- Trained video-conditioned robot policies with cross-attention transformers. [Vid2Robot \(RSS'24\)](#)
 - Extended models for visual grounding with conditional caption generation for QA. [FlexCap \(NeurIPS'24\)](#)
 - Evaluated LLM prompts for low-level motion control for articulated object interaction. [PromptBook \(ICRA'24\)](#)
 - Developed pipelines for zero-shot and fine-tuned task evaluations for single neural network policy trained on aggregated robotics data. [Open X-Embodiment \(ICRA'24\) Best Paper.](#)
- Sep'21– **Meta (Facebook), Menlo Park, CA, USA.**
- Aug'22 AI resident with [Akshara Rai](#) and [Yixin Lin](#)
- Developed synthetic data for robot demonstrations of loading the simulated dishwasher in AI Habitat. [Code](#)
 - Trained Encoder-decoder Transformer-based high-level policy for learning preferences in loading dishwasher from a single demonstration as prompt. [Paper](#) | [Website](#)
- Aug'18– **Microsoft Research, Bengaluru, India.**
- Aug'19 Research Fellow with [Amit Deshpande](#) and [Navin Goyal](#)
- **Divergence minimization in GANs** Investigated the properties of Jensen Shannon and Wasserstein divergences between the given and the generated distributions during training neural network generators, to discuss the learning dynamics like convergence and generalization.
- Jan'18– **Montreal Institute for Learning Algorithms (MILA), Montreal, QC, Canada.**
- Jun'18 Research Intern with [Aaron Courville](#)
- **Out-of-Distribution Detection in Generative Models** Experimented on autoregressive generative models and Variational Autoencoders, to detect any 'out-of-distribution (OOD)' sample at deployment. [Thesis](#)
- May'17– **Simon Fraser University (SFU), Burnaby, BC, Canada.**
- Jul'17 MITACS Globalink Research Intern with [Oliver Schulte](#)
- **Learning Bayesian Networks for Relational Databases** Remodelled existing codebase for project *FactorBase* to integrate cross-table sufficient statistics or population variables in the contingency tables for learning ground-level sufficient statistics. [Code](#) | [Invited Paper](#)

- May'16– **Forschungszentrum Informatik (FZI)**, Karlsruhe, Germany.
- Jul'16 Research Intern with [York Sure-Vetter](#) and [Matthias Frank](#)
- **API annotation platform using Semantic Web** Designed and developed a customized Semantic MediaWiki for the decision support system of Project *BigGIS*; [Extended Abstract](#) | [Poster](#)

Selected Publications

- Aug 2024 **ANAVI: Audio Noise Awareness using Visuals of Indoors for NAVigation.**
Vidhi Jain, Rishi Veerapanini, Yonatan Bisk. *In Proceedings of Conference of Robot Learning (CoRL) 2024*, Munich, Germany. [Paper](#) | [Website](#)
- Mar 2024 **FlexCap: Generating Rich, Localized, and Flexible Captions in Images.**
Debidatta Dwibedi, **Vidhi Jain**, Jonathan Tompson, Andrew Zisserman, Yusuf Aytar. *Accepted at Conference on Neural Information Processing Systems (NeurIPS) 2024*, Vancouver, Canada. [Paper](#) | [Website](#)
- Jan 2024 **Vid2Robot: End-to-end Video-conditioned Policy Learning with Cross-Attention Transformers.**
Vidhi Jain, Maria Attarian, Nikhil J Joshi, Ayzaan Wahid, Danny Driess, Quan Vuong, Pannag R Sanketi, Pierre Sermanet, Stefan Welker, Christine Chan, Igor Gilitschenski, Yonatan Bisk, Debidatta Dwibedi, *In Proceedings of Robotics: Science and Systems (RSS) 2024*, Delft, Netherlands. [Paper](#) | [Website](#) | [Video](#)
- Nov 2023 **How to Prompt Your Robot: A PromptBook for Manipulation Skills with Code as Policies.**
Montserrat Gonzalez Arenas, Ted Xiao, Sumeet Singh, **Vidhi Jain**, Allen Z. Ren, Quan Vuong, Jake Varley, Alexander Herzog, Isabel Leal, Sean Kirmani, Dorsa Sadigh, Vikas Sindhwani, Kanishka Rao, Jacky Liang, Andy Zeng. *In Proceedings at International Conference on Robotics and Automation (ICRA) 2024*, Yokohama, Japan. [Paper](#)
- Nov 2023 **Open X-Embodiment: Robotic Learning Datasets and RT-X Models.**
Open X-embodiment collaboration. *In Proceedings at International Conference on Robotics and Automation (ICRA) 2024*, Yokohama, Japan. [Best Paper Paper](#) | [Website](#)
- Jun 2023 **SLAP: Spatial-Language Attention Policies.**
Priyam Parasher, **Vidhi Jain**, Xiaohan Zhang, Jay Vakil, Sam Powers, Yonatan Bisk, Chris Paxton. *In Proceedings of Conference on Robot Learning (CoRL) 2023*, Atlanta, USA., [Paper](#) | [Website](#)
- Jun 2023 **HomeRobot: Open-Vocabulary Mobile Manipulation.**
Sriram Yenamandra, Arun Ramachandran, Karmesh Yadav, Austin S Wang, Mukul Khanna, Theophile Gervet, Tsung-Yen Yang, **Vidhi Jain**, Alexander Clegg, John M Turner, Zsolt Kira, Manolis Savva, Angel X Chang, Devendra Singh Chaplot, Dhruv Batra, Roozbeh Mottaghi, Yonatan Bisk, Chris Paxton. *In Proceedings of Conference on Robot Learning (CoRL) 2023*, Atlanta, USA. [Paper](#) | [Website](#) | [Competition @ NeurIPS 2023](#)
- Dec 2022 **Transformers are Adaptable Task Planners.**
Vidhi Jain, Yixin Lin, Eric Undersander, Yonatan Bisk, Akshara Rai. *In Proceedings of Conference on Robot Learning (CoRL) 2022*, Auckland, New Zealand. [Paper](#) | [Website](#) | [Video](#) | [Code](#)

Workshop Presentations

- Dec 2022 **MAEA: Multimodal Attribution for Embodied AI.**
Vidhi Jain, Jayant Sravan Tamarapalli, Sahiti Yerramilli, Yonatan Bisk. *Presented at NeurIPS 2022, 5th Robot Learning Workshop: Trustworthy Robotics (RLW 2022)*, New Orleans, USA. [Short Paper](#) | [Website](#) | [Video](#)
- Oct 2020 **Predicting Strategies in Simulated Search and Rescue.**
Vidhi Jain, Rohit Jena, Huao Li, Tejus Gupta, Dana Hughes, Michael Lewis, Katia Sycara. *Presented at NeurIPS AI+HADR 2020*, Virtual. [Video](#) | [Preprint](#)
- Oct 2020 **Learning Embeddings that Capture Spatial Semantics for Indoor Navigation.**
Vidhi Jain, Shishir Patil, Prakhar Agarwal, Katia Sycara. *Presented at NeurIPS 2020 Object Representations for Learning and Reasoning (ORLR 2020)*, Virtual. [Video](#) | [Preprint](#) | [Code](#)
- Sep 2020 **Learning to Navigate in Unseen Cluttered Structured Environments.**
Vidhi Jain, Ganesh Iyer, Katia Sycara. *NeurIPS 2020 Women In Machine Learning (WiML 2020)*, Virtual. [Poster](#)

- Jul 2020 **Coping with Sample Inefficiency in Deep Reinforcement Learning (DRL) in embodied AI.**
Vidhi Jain, Simin Liu, Ganesh Iyer. ICML'20 Women In Machine Learning (WiML) Un-Workshop, Virtual.
[Discussion](#) | [Presentation](#)
- Aug 2017 **Model Selection Scores for Multi-Relational Bayesian Networks.**
 Sajjad Gholami, Oliver Schulte, **Vidhi Jain**, Qiang Zhao. IJCAI 2017 Declarative Learning Based Programming (DeLBP). [Invited Paper](#)

Preprints

- Dec 2023 **Towards General-Purpose Robots via Foundation Models: A Survey and Meta-Analysis.**
 Yafei Hu*, Quanting Xie*, **Vidhi Jain***, Jonathan Francis, Jay Patrikar, Nikhil Keetha, Seungchan Kim, Yaqi Xie, Tianyi Zhang, Shibo Zhao, Yu Quan Chong, Chen Wang, Katia Sycara, Matthew Johnson-Roberson, Dhruv Batra, Xiaolong Wang, Sebastian Scherer, Zsolt Kira, Fei Xia, Yonatan Bisk. [Paper](#) | [Website](#)
- Nov 2019 **Towards Zero-Shot Alignment and Retrieval for Forensic Detection.**
Vidhi Jain*, Amrit Setlur*, Aditya Pratapa*. [Report](#)

Tech Reports

- Aug 2025 **Lowering Barriers in Human-Robot Communication.** PhD Thesis. [Report](#)
- Aug 2021 **Towards Explainable Embodied AI.** Masters Thesis. [Report](#)
- Jun 2018 **Investigating the viability of Generative Models for Novelty Detection.** Bachelors Thesis. [Report](#)

Selected Awards and Honors

- Dec 2025 **BITS Global 30 Under 30 Award** in Research leaders – Academia category [Link](#)
- Aug 2020 **CIFAR Deep Learning Reinforcement Learning Summer School**
- Apr 2020 **Heidelberg Laureate Forum (HLF '20)** selected as a Young Researcher among 224 others worldwide.
- Jul 2019 **J N Tata Endowment Scholarship for Higher Studies, India**
- Jun 2019 **K. C. Mahindra Scholarships for Post-Graduate Studies Abroad**
- May 2017 **ACM-W Scholarship for Attendance at Research Conferences**
- Mar 2017 **Citi Women Leader Award (CWLA), Mumbai, India**, Awarded one year of study scholarship covering upto INR 400,000 i.e. USD 6000, among the top 3 out of 1200 applicants
- Jun 2016 **GE Foundation Scholar-Leaders Program (GEFSLP)**, Awarded USD 2000 for two years of study, among top 5 scholars selected in India
- Mar 2016 **National Finalists at Microsoft's 'Build the Shield', Hyderabad, India**, Ranked 15th nationwide
- 2014–15 **Merit Scholarship** by Dean, BITS Pilani, *Top 1% among the batch of about 900 students*
- 2013 **Awarded Kishore Vaigyanik Protsahan Yojna (KVPY) Fellowship** by Dept of Science and Tech., Govt. of India
- 2010 **Awarded National Talent Search Exam (NTSE) Scholarship** by NCERT, India
- 2010 Ranked Top 1% in Regional State in **National Standard Examination in Junior Science (NSEJS)**

Talks

- Mar 2024 ***How to think of multimodality in Embodied AI and Robotic Manipulation?***
 Guest Lecture in Multimodal Machine Learning course. [Lecture Video](#)
- Apr 2023 ***Training vision models in PyTorch.***
 Hands-on lecture in Computer Vision course. [Lecture Video](#)
- Oct 2021 ***Three brushes to paint your research canvas.***
 SAIDL (Virtual) [Video](#) | [Event Poster](#)

- May 2019 **Tutorial on Deep Learning with PyTorch.**
PyLadies Bangalore, India. [Presentation](#) | [Event Poster](#)
- Jan 2019 **The One in Asankhya Project.**
by Sukriti Paul and Mansi Goyal (Virtual) [Video](#) | [Blog](#)
- Jul 2017 **Invited panelist for 'Innovation by Young India'.**
telecasted on national news channel NDTV India at New Delhi office. [Video](#)

Professional Service and Leadership

- Sep'23– Present **Reviewer** at IEEE International Conference on Robotics and Automation (ICRA), International Conference on Intelligent Robots and Systems (IROS)
- Jan'20– Present **Reviewer** at International Conference on Machine Learning (ICML), Neural Information Processing and Signal (NeurIPS) and International Conference on Learning Representations (ICLR).
- Jun'20– **Workshop Session Organizer** | Coping with Sample Inefficiency in Deep Reinforcement Learning (DRL) in embodied AI, ICML'20 WiML Un-Workshop, Virtual. [Report](#) | [Presentation](#)
- Sep'17– **Co-Founder and Vice Chair** | BITS Pilani Association for Computing Machinery-Women (ACM-W) Chapter, *Started with 16 on-campus girls, organized project brainstorming and mentoring sessions*
- May'15– **Technical Volunteer** | ARISE Impact, non-profit organization *Developed Android application for Audio-based learning for visually challenged*

Independent Research Supervision and Mentorship

- Feb 2024 **Soumya Teotia** (*Senior thesis + intern*) Multi-task instruction following in Habitat simulator and developing code for easy transfer to physical robot. → **Masters at CMU RI**
- May 2024 **Su Li** (*CMU sophomore*) Social teleoperation with VR headset and Hello Robot Stretch. Presented live demo at CoRL 2024, Munich, Germany. → **Intern at MathWorks**
- Aug 2024 **Ashish Marisetty** (*CMU MIIS graduate for independent study*) Language-conditioned video generation for high-level robot dynamics modeling. → **Intern at Nvidia**

Teaching

- Jan'24– May'24 **Teaching Assistant** | 11777 Multimodal Machine Learning by Yonatan Bisk. Mentoring 5 teams with 3-5 students each for semester-long research projects.
- Jan'23– May'23 **Teaching Assistant** | 16720 Computer Vision by Deva Ramanan. for 106 students enrolled in the course, prepared homework, and graded.
- Jan'17– May'17 **Teaching Assistant** | CS F111 Computer Programming by Vishal Gupta, *Mentored 35 students for 4 hours per week for Code programming lab sessions.*
- Aug'14– May'15 **Teaching Volunteer** | Computer Literacy Programme, National Service Scheme, *Tutored 12-15 rural natives for 3 hours per week on computer applications.*